

Completeness $\wedge$ Transitivity $\wedge$ Independence $\wedge$ Mixture continuity $\Rightarrow$ Expected-utility representation

Result — human; produced by literature.

0.1 Premises

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \succcurlyeq r\}$ and $\{\lambda \in [0,1] : r \succcurlyeq \lambda p + (1-\lambda)q\}$ are closed.

0.2 Conclusion

- **Expected-utility representation.** There is a function $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$.

0.3 Proof

The von Neumann–Morgenstern theorem in the Herstein–Milnor mixture-set form. Continuity plus weak order give, for any $p \succcurlyeq q \succcurlyeq r$ with $p \succ r$, a unique λ with $q \sim \lambda p + (1-\lambda)r$ (connectedness of $[0,1]$ and closedness of both sections). Fix a best and a worst lottery (exist since X is finite and $\succcurlyeq$ is a weak order respecting mixtures), define $U(q)$ as that λ , and use independence to show U is affine in mixtures; then $u(x) = U(\delta_x)$ represents $\succcurlyeq$ by expected utility.

0.4 Sources

- von Neumann & Morgenstern (1944), Theory of Games and Economic Behavior, appendix
- Herstein & Milnor (1953), An axiomatic approach to measurable utility, Econometrica 21, Theorem 8
- Kreps (1988), Notes on the Theory of Choice, Theorem 5.15

Record: topics/decision-theory/results/vnm-theorem.yaml